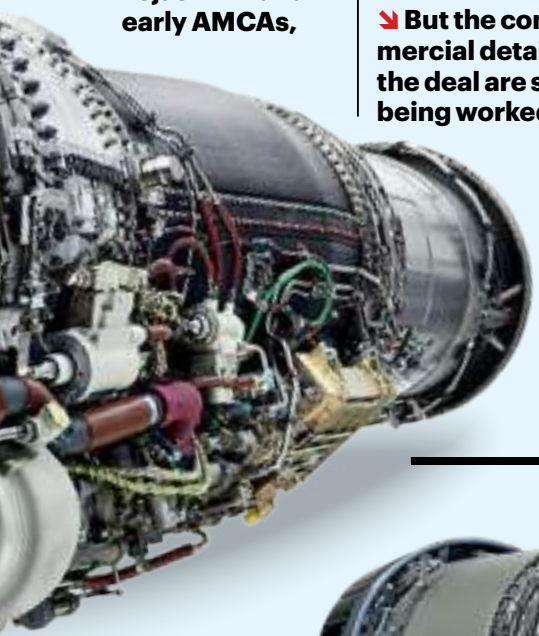
A JET ENGINE FACE-OFF

General Electric's F414 is the chosen engine for Tejas Mk 2 and the first batch of AMCA. Now, the later AMCAs will have an engine co-developed with Safran. What do the two firms offer India, and at what stage are the deals?

GENERAL ELECTRIC

- Under the 2023 agreement, the *GE F414*, meant for Tejas Mk 2 and early AMCAs, will be made in India with 80% transfer of technology
- But the commercial details of the deal are still being worked out
- The F414 is an afterburning turbofan engine with 97.9 kN thrust
- The company is also to deliver 99 GE F404 engines for the Tejas Mk 1A fighter under a 2021 deal worth Rs 5,375 cr
- It has sent only four F404s in 2025, attracting a penalty for delayed delivery by India





SAFRAN

- DRDO/ HAL and Safran plan to co-develop a jet engine of 120-140 kN class for the advanced version of the stealth AMCA, to be in production by 2035
- The proposal is awaiting inter-ministerial clearances before the Cabinet Committee on Security gives final approval
- The plan is for 100% transfer of tech for around 100 engines, all parts to be India-made
- The engine will be a more powerful version of the *M-88* used in Rafale
- Safran has also developed the Shakti helicopter engine with HAL
- Shakti also promised to include all Indian parts, but less then 50 per cent are fully indigenous

is good reason for this. Engines for the other two indigenous fighter jets—the Light Combat Aircraft (LCA) Tejas Mk 1A and the LCA Tejas Mk 2—are tied to engines manufactured by GE, and progress here has been unsatisfactory. GE engine delivery is running behind its estimated timelines and has hampered the Tejas programme. HAL is in the process of procuring GE F404-IN20 engines from GE Aerospace for its Tejas Mk 1A fighter jets, with a Rs 5,375 crore deal for 99 engines signed in 2021 and a follow-on $1 billion (Rs 8,559 crore) deal for 113 engines expected to be signed soon. Though delivery was to start in March 2023, GE Aerospace has delivered the first four engines this year.

₹62,149 CRORE

Or $7 billion—estimated cost of the planned deal between Safran and DRDO/ HAL to make jet engines for the 5th generation AMCA

Consequently, not a single Mk 1A of the initial 83 ordered by the Indian Air Force has been delivered. GE blamed global supply chain disruptions for the delay, which has forced India to impose penalties on the firm.

Simultaneously, commercial aspects of the much-vaunted $1.5 billion (Rs 13,317 crore) deal between HAL and GE for co-production of the